
Data Discovery and Management Tool - 'appdataexplorer'

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APPDATAEXPLORER

1.1 config module

This module contains the core structural elements of the data discovery tool. The core structural elements consists of the names of applications, HPC centers, ESMs etc. The names of the file types produced by each application and their extensions. Apart from that certain data structures to hold these infos.

class config.configAppdataexplorer

Bases: object

Class to hold the various paramters to drive the appdataexplorer.

activeConfig = None

appDataSrcNameFileExt

List of the file extensions.

Type

list[str]

appDataSrcNames

List of the file types. Names of the different types of data sources / files the applications can produce.

Type

list[str]

appDescriptionInfo

Dictionary of the description for each of the apps. Data structure to store the description, provider etc corresponding to each app.

Type

dict[str]

appNames

List of the application names. Names of the applications using the GSV data and producing respective application specific data.

Type

list[str]

dataPath

Path to location of the data produced by the applications.

Type

str

esmNames

List of the ESMs. Names of the Earth System Models used performing the simulations and producing the GSV data.

Type

list[str]

getDataPath()

Get the application data path.

For setting the location where the apps store the data that is to be cataloged by the appdataexplorer.

Parameters

None. –

Returns

Path to location of the data produced by the apps.

Return type

path

getESMs()

Set the ESM names.

For getting the ESM names from the configuration.

Parameters

None. –

Returns

List of the ESM names.

Return type

list[str]

getHPCCenters()

Get the HPC centers names.

For getting the HPC center names from the configuration.

Parameters

None. –

Returns

List of the HPC center names.

Return type

list[str]

getInstallationPath()

Get the appdataexplorer installation path.

For getting the location where the appdataexplorer is intalled on the disk.

Parameters

None. –

Returns

Path to appdataexplorer.

Return type

str

getOutputPath()

Get the output path.

For getting the location where the catalogs produced by appdataexplorer will be stored.

Parameters

None. –

Returns

Path to location of the catalogs produced by the appdataexplorer.

Return type

path

getappDataSrcNameFileExt()

Get the file name extension for the data sources.

For getting the file name extensions,for the various types of data sources that the data produced by the apps belongs to, from the the configuration.

Parameters

None. –

Returns

Dictionary of the data source names and the corresponding file name extensions.

Return type

appDataSrcNameFileExt

getappDataSrcNames()

Get the data source names.

For getting the data source names like 'netcdf', 'image', 'text' from the configuration.

Parameters

None. –

Returns

List of the data source names.

Return type

list[str]

getappDataSrcs(*app*)

Get the data sources produced by a given application.

For getting the data sources to which the data produced by a given application belongs to, from the configuration.

Parameters

None. –

Returns

List of data source names.

Return type

appDescInfo

getappDescription(*app*)

Get the description of a given application.

For getting the description about a given application from the configuration.

Parameters

app – Application name.

Returns

Application description.

Return type

str

getappDescriptionInfo()

Get the application description from the configuration.

For getting the application description info, provider etc from the configuration.

Parameters

None. –

Returns

Dictionary of the various parameters that describe the various apps.

Return type

appDescInfo

getappMetadataKeys(*app*)

Get the metadata keys for a given application.

For getting the metadata keys for the data produced by a given application from the configuration.

Parameters

app – Application name.

Returns

List of metadata keys.

Return type

str

getappNames()

Get the application names.

For getting the application names from the configuration.

Parameters

None. –

Returns

List of the application names.

Return type

list[str]

getappProvider(*app*)

Get the provider of a given application.

For getting the details about the provider of a given application from the configuration.

Parameters

app – Application name.

Returns

Details of the application provider.

Return type

str

hpcCenters

List of the HPC centers. Names of the HPC centers producing the data.

Type

list[str]

installationPath

Path to location of the data discovery and management tool on disk.

Type

str

outputPath

Path to location of the catalog produced by data discovery and management tool.

Type

str

setConfigfromJSON(*jsonDict*)

Set the appdataexplorer configuration to the input JSON dictionary.

For setting the configuration parameters of the appdataexplorer from the input JSON dictionary.

Parameters

jsonDict – A JSON dictionary.

Returns

None.

setDataPath(*path*)

Set the application data path.

For setting the location where the apps store the data that is to be cataloged by the appdataexplorer.

Parameters

path – Path to location of the data produced by the apps.

Returns

None.

setESMs(*esmNames*)

Set the ESM names.

For updating the configuration with the ESM names.

Parameters

list[str] – List of the ESM names.

Returns

None.

setHPCCenters(*hpcCenters*)

Set the HPC centers names.

For updating the configuration with the HPC center names.

Parameters

list[str] – List of the HPC center names.

Returns

None.

setInstallationPath(*path*)

Set the appdataexplorer installation path.

For setting the location where the appdataexplorer is intalled on the disk to the configuration.

Parameters

path – Path to appdataexplorer.

Returns

None.

setOutputPath(*path*)

Set the output path.

For setting the location where the catalogs produced by appdataexplorer will be stored.

Parameters

path – Path to location of the catalogs produced by the appdataexplorer.

Returns

None.

setappDataSrcNameFileExt(*appDataSrcNameFileExt*)

Set the file name extension for the data sources.

For updating the configuration with the file name extensions for the various types of data sources that the data produced by the apps belongs to.

Parameters

appDataSrcNameFileExt – Dictionary of the data source names and the corresponding file name extensions.

Returns

None.

setappDataSrcNames(*appDataSrcNames*)

Set the data source names.

For updating the configuration with the data source names like 'netcdf', 'image', 'text'.

Parameters

list[str] – List of the data source names.

Returns

None.

setappDescriptionInfo(*appDescInfo*)

Set the application description info to the configuration.

For updating the configuration with the application description info, provider etc.

Parameters

appDescInfo – Dictionary of the various parameters that describe the various apps.

Returns

None.

setappNames(*appNamesList*)

Set the application names.

For updating the configuration with the application names.

Parameters

list[str] – List of the application names.

Returns

None.

`config.initializeAppdataexplorer(configFile)`

Initializing the appdataexplorer.

Initialize the appdataexplorer library configuration with the parameters provided in the input json file.

Parameters

configFile – JSON file with configuration parameters.

Returns:

curconfig - current configuration object.

1.2 intakeCuration module

This module contains the APIs for updating the intake catalog for each app and each of it's file types. These APIs need to be called by the app data curator for maintaining the intake catalog.

`intakeCuration.createAppIntakeCatalog()`

Create yaml files for the Apps.

For each of the Apps, create the yaml files containing the links to the yaml files for the ESMs .

Parameters

None –

Returns

None

`intakeCuration.createESMIntakeCatalog()`

Create yaml files for the ESMs.

For each of the ESMs, create the yaml files containing the links to the yaml files for the available sources .

Parameters

None –

Returns

None

`intakeCuration.createHPCIntakeCatalog()`

Create yaml files for the HPC centers.

For each of the HPC centers, create the yaml files containing the links to the yaml files for the Apps .

Parameters

None –

Returns

None

`intakeCuration.createIntakeCatalogSourcesForFileList(app, src, catFile, srcFileList)`

Create catalog for the sources for the inout file list.

For all the source files, create the intake counterparts and write out to the catalog file.

Parameters

- **app** – application Name
- **src** – source Name

- **catFile** – Catalog File Name.
- **srcFileList** – List of netcdf files.

Returns

None

`intakeCuration.createMasterIntakeCatalog(catalogFileName)`

Create master catalog file for the HPC centers.

Create the top level master catalog containing the links to the yaml files for the HPC centers.

Parameters

catalogFileName – String to hold the name of top level catalog file that would be created.

Returns

None

`intakeCuration.createSrcIntakeCatalog()`

Create yaml files for the sources.

For each of the app, check the available sources (file types) and create the corresponding yaml files.

Parameters

None –

Returns

None

`intakeCuration.intakeSrcCreator(app, src, srcFile, appMetadataKeys)`

Create yaml resource for the input source file.

For the input file of given source and app and metadata keys, prepare the intake object.

Parameters

- **app** – application Name
- **src** – source Name
- **catFile** – Catalog File Name.
- **srcFileList** – List of netcdf files.

Returns

intake catalog object.

Return type

intakeSrc

1.3 stacCuration module

This module contains the APIs for updating the STAC catalog for each app and each of its file types. These APIs need to be called by the app data curator for maintaining the STAC catalog.

`stacCuration.createAppSTACCatalog()`

Create STAC catalog for the Apps.

For each of the Apps, create the STAC catalog containing the links to the STAC catalogs for the ESMs.

Parameters

None. –

Returns

None.

`stacCuration.createESMSTACCatalog()`

Create STAC catalog for the ESMs.

For each of the ESMs, create the STAC catalog containing the links to the STAC catalogs for the available sources.

Parameters

None. –

Returns

None.

`stacCuration.createHPCSTACCatalog()`

Create STAC catalog for the HPC centers.

For each of the HPC centers, create the STAC catalog containing the links to the STAC catalogs for the Apps.

Parameters

None. –

Returns

None.

`stacCuration.createMasterSTACCatalog()`

Create master STAC catalog file for the HPC centers.

Create the top level master STAC catalog containing the links to the STAC catalogs for the HPC centers.

Parameters

None. –

Returns

None.

`stacCuration.createSTACSourcesForFileList(app, src, srcCatalog, srcFileList)`

Create STAC items for the netcdf files in the input

file list of files.

For each of the netcdf files in the input file list, create the STAC item and save in the input catalog file.

Parameters

- **app** – Application Name.
- **src** – Source Name.
- **catFile** – Catalog File Name.
- **srcFileList** – List of netcdf files.

Returns

None

`stacCuration.stacItemCreator(app, src, srcFile, appMetadataKeys, itemId)`

Create stac item for the input source file.

For the input file of given source and app and metadata keys, prepare the STAC item.

Parameters

- **app** – application Name
- **src** – source Name

- **catFile** – Catalog File Name.
- **srcFileList** – List of netcdf files.

Returns

intake catalog object.

Return type

intakeSrc

1.4 utils module

This module contains certain utility functions like for fetching the files from a given location on disk etc.

utils.getAppSrcFileList(*app, src, esm, srcExtList*)

Create a list of files produced by the app, for a given file type.

Search the app data directory and create a list of all the files of a given file type like netcdf, jpeg etc.

Parameters

- **app** – Application Name.
- **src** – Source or file type name like netcdf, image, text.
- **esm** – Earth system model Name.
- **srcExtList** – The file suffix like *.nc, *.jpg, *.pdf, *.txt, *.csv.

Returns

srcFileList - a list.

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